



INDIAN INSTITUTE OF INFORMATION TECHNOLOGY DESIGN AND MANUFACTURING KURNOOL

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Jagannathagattu, Dinnedevaram, Kurnool, Andhra Pradesh — 518008

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CLOUDCAMPUS

College Campus Management System

AWS MULTI-TIER CLOUD DEPLOYMENT & ENGINEERING REPORT

Course: Cloud Computing • Academic Session: 2026–2027

PROJECT DEVELOPED & SUBMITTED BY:

Chakala Karthik • Roll No: 123CS0038

Shaik Venkat • Roll No: 123CS0037

B.Tech Computer Science & Engineering

COURSE INSTRUCTOR & PROJECT GUIDE:

Dr. Anil Kumar

Assistant Professor

Department of Computer Science & Engineering

IIITDM Kurnool

1. Executive Summary & Project Abstract

CloudCampus is an enterprise-grade, multi-tier College Campus Management System architected, deployed, and verified entirely on **Amazon Web Services (AWS)** in the us-east-1 (N. Virginia) region. The system transitions traditional monolithic campus administration into a high-availability, secure, and observable cloud platform serving three distinct academic roles: **Students, Faculty, and Administrators**.

Primary Engineering Thesis: Rather than using cloud infrastructure solely for basic virtual machine hosting, CloudCampus deeply integrates **14 native AWS services**—mapping each architectural responsibility (Identity, Ingress, Routing, Compute, Managed Database, Object Storage, Event-Driven Serverless, Message Queuing, Scheduled Automation, Secrets Governance, Access Control, Auditing, Network Isolation, and Observability) to specialized cloud primitives.

1.1 Core Architectural Highlights

- **Serverless Cloud Identity:** Complete deprecation of local database passwords in favor of **AWS Cognito User Pools** (us-east-1_Ic9huqJjL) with Cognito Groups and cryptographically verified JWT tokens.
- **Ingress & Private Routing:** Zero direct public exposure of backend servers; external traffic enters via **AWS API Gateway HTTP API** (7k2yo6gy77) and routes across an **AWS Application Load Balancer (ALB)** into private EC2 subnets.
- **Dual Database & Storage Separation:** Relational institutional data resides in **Amazon RDS PostgreSQL** (cloudcampus-db) while unstructured binary artifacts and profile images are stored directly in **Amazon S3** (cloudcampus-511225358997).
- **Asynchronous Serverless Workflows:** Faculty assignment creation asynchronously triggers **AWS Lambda** (CloudCampus-Assignment-Notification) to identify enrolled students and dispatch alerts via **Amazon SNS**; scheduled deadlines are tracked by **Amazon EventBridge** triggering reminder Lambdas with duplicate prevention.
- **Full-Stack Observability & Auditing:** Native **AWS CloudWatch Monitoring Dashboard** (CloudCampus-Monitoring) tracking 12 live telemetry metrics across EC2, ALB, RDS, and Lambda alongside immutable RDS audit logging.

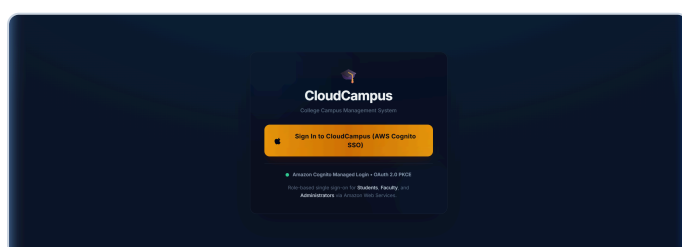


Figure 1.1 — CloudCampus Cloud-Only Authentication Interface

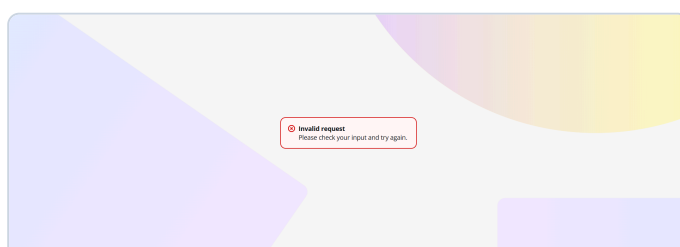


Figure 1.2 — AWS Cognito Hosted UI & Identity Provider

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3. AWS Services Used in CloudCampus

CloudCampus leverages a comprehensive suite of AWS services divided into **Runtime Application Services** (direct participants in user requests and asynchronous jobs) and **AWS Management / Development Services** (used for administrative deployment and governance).

AWS SERVICE	CLASSIFICATION	SPECIFIC ARCHITECTURAL PURPOSE IN CLOUDCAMPUS
Amazon EC2	RUNTIME	Hosts the clustered Express + Prisma application server (i-03681025582d882c5) on Amazon Linux 2023 under PM2.
Amazon API Gateway	RUNTIME	HTTP API (7k2yo6gy77) providing public HTTPS entry, CORS headers, edge throttling, and direct ALB forwarding.
AWS Cognito	RUNTIME	User Pool (us-east-1_Ic9huqJjL) managing user identities, password authentication, OAuth2 OIDC tokens, and RBAC groups.
AWS ALB	RUNTIME	Application Load Balancer distributing HTTP traffic across target instances with automated /health health checking.
Amazon RDS PostgreSQL	RUNTIME	Primary relational database (cloudcampus-db) running PostgreSQL 17.5 in private database subnets with SSL encryption.
Amazon S3	RUNTIME	Encrypted object storage bucket (cloudcampus-511225358997) for user profile photos and assignment submission files.
AWS Secrets Manager	RUNTIME	Secure storage of PostgreSQL connection credentials (cloudcampus/rds); injected via IAM at runtime without hardcoding.
AWS Lambda	RUNTIME	In-VPC serverless functions for event-driven assignment notification fanout and deadline reminder verification.
Amazon SNS	RUNTIME	Publish/subscribe topic (CloudCampus-Notifications) broadcasting academic notifications to enrolled students.
Amazon EventBridge	RUNTIME	Automated scheduled cron rule (rate(1 day)) triggering daily assignment due-date reminder scans.
Amazon CloudWatch	RUNTIME	12-widget monitoring dashboard (CloudCampus-Monitoring), 4 metric alarms, and application log stream aggregation.
Amazon VPC	RUNTIME	Network isolation boundary (vpc-0146f9a06bf1163a6) with private subnets, security groups, and Interface Endpoints.
AWS IAM	RUNTIME	Least-privilege execution roles (CloudCampus-EC2-Role, CloudCampus-Lambda-Execution-Role) and security policies.
AWS CloudShell	MANAGEMENT	Browser-based CLI environment used for deployment, database migrations, and AWS CLI verification (not in runtime path).
Billing & Cost Mgmt	MANAGEMENT	Cost allocation, budget tracking, and Free Tier consumption governance across provisioned project resources.

Database Engine Clarification: While the AWS console groups relational engines under “Aurora and RDS”, CloudCampus specifically deploys **Amazon RDS for PostgreSQL** (v17.5).

4. AWS Service Categorization & Architecture Roles

The table below categorizes the deployed AWS cloud services according to foundational cloud computing domains, highlighting the separation of concerns across compute, networking, data persistence, and governance.

FUNCTIONAL DOMAIN	AWS SERVICES UTILIZED	ARCHITECTURAL ROLE & SCOPE
Compute & Processing	Amazon EC2, AWS Lambda	Stateless REST API backend cluster and event-driven serverless background workers.
Networking & Ingress	Amazon VPC, AWS ALB, Amazon API Gateway	Network perimeter isolation, edge SSL termination, Layer-7 load balancing, and routing.
Relational Persistence	Amazon RDS for PostgreSQL	ACID-compliant relational database for academic records, enrollments, and audit logs.
Object Storage	Amazon S3	High-durability storage for user avatars, assignment PDFs, and submission archives.
Identity & Security	AWS Cognito, AWS IAM, AWS Secrets Manager	OIDC user authentication, RBAC group tokens, least-privilege IAM, and runtime secret injection.
Messaging & Scheduling	Amazon SNS, Amazon EventBridge	Asynchronous pub/sub notification delivery and scheduled cron automation.
Observability & Alarms	Amazon CloudWatch	Live multi-service telemetry dashboard, metric threshold alarms, and log stream collection.
Cloud Administration	AWS CloudShell, AWS Billing & Cost Mgmt	Browser-based CLI administration, SSM remote execution, and cost budget monitoring.

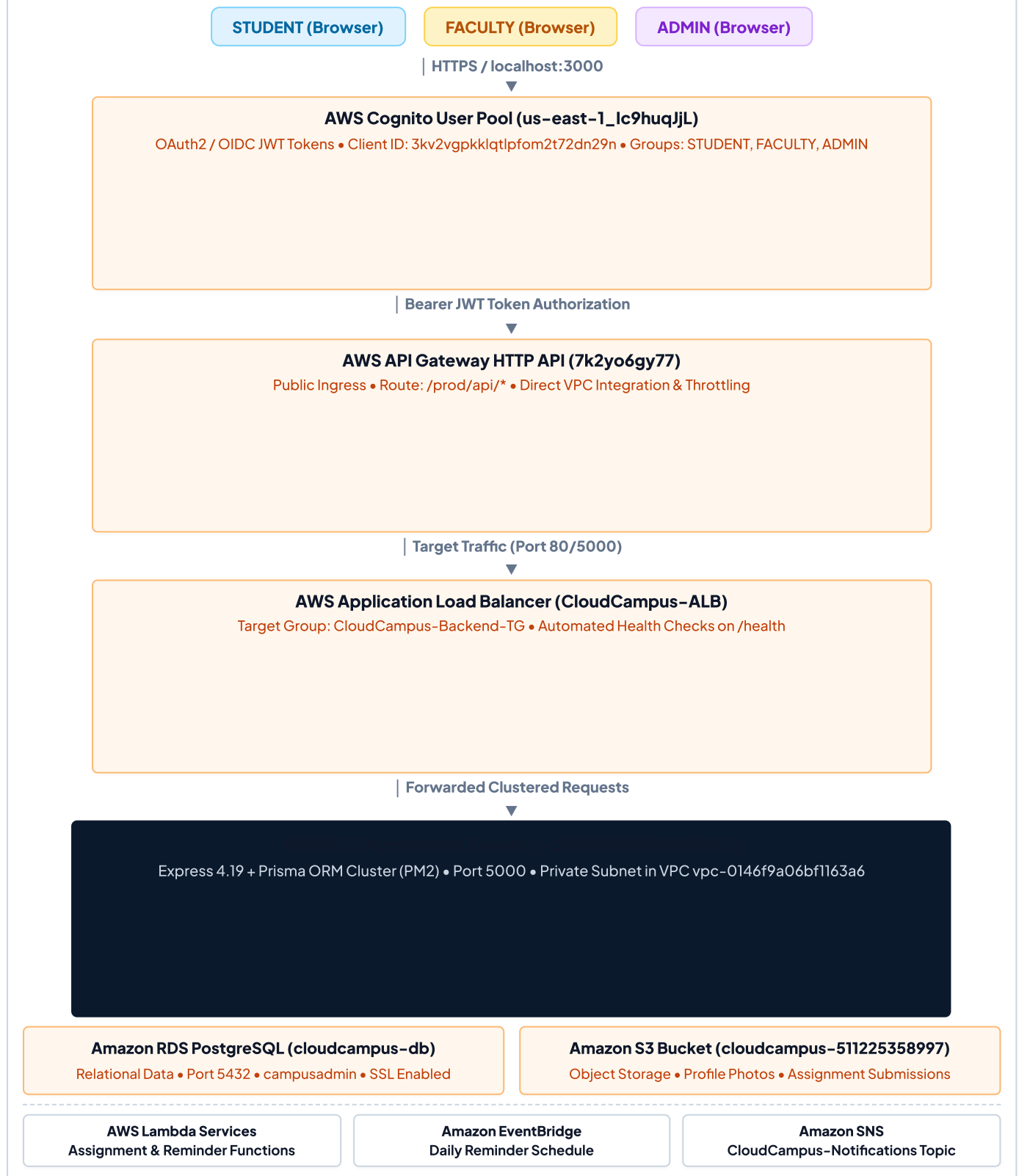
Service Responsibility & Integration Pipelines

1. Request Ingress & Backend Compute Pipeline: Client Browser (HTTPS) → AWS Cognito JWT → API Gateway (7k2yo6gy77) → AWS ALB → EC2 Express Backend (Port 5000) → RDS PostgreSQL / S3.	2. Asynchronous Serverless & Monitoring Pipeline: Faculty Action → EC2 Lambda Invoke → In-VPC Lambda → Amazon SNS Broadcast • Amazon EventBridge Cron → Reminder Lambda • CloudWatch Telemetry.
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5. Complete System Cloud Architecture

The diagram below illustrates the end-to-end multi-tier cloud architecture of CloudCampus. Requests originate from client browsers over HTTPS, authenticate with AWS Cognito, enter through AWS API Gateway, balance across the Application Load Balancer, execute on the clustered EC2 compute instance, persist in Amazon RDS PostgreSQL and Amazon S3, and trigger asynchronous AWS Lambda functions and CloudWatch telemetry.

FIGURE 5.1 — FULL-PAGE MULTI-TIER CLOUD DEPLOYMENT ARCHITECTURE DIAGRAM



Decoupled Cloud Primitives: User authentication, public ingress, application compute, relational persistence, and object storage operate as isolated managed tiers.

6. Amazon VPC Network Boundary & Subnets

All compute, database, and serverless Lambda executions are enclosed inside **Amazon Virtual Private Cloud (VPC)** vpc-0146f9a06bf1163a6 across availability zones in us-east-1.

Figure 6.1 — Amazon VPC Network Isolation: vpc-0146f9a06bf1163a6 (CIDR: 10.0.0.0/16)

Public / Ingress Subnets

- subnet-030b656a0b401b23d (us-east-1a)
 - subnet-0b8c7db479db1b113 (us-east-1b)
- Hosts: AWS Application Load Balancer (ALB)

Private Compute & DB Subnets

- subnet-0ea7b2a7ac8952aa9 (us-east-1a)
 - subnet-02f2f01a92b63d057 (us-east-1b)
- Hosts: EC2 Backend, RDS PostgreSQL, VPC Lambdas

VPC Interface & Gateway Endpoints

S3 Gateway:

vpce-0abe4018a7179df71

Secrets Manager:

vpce-03453d022c9090b54

Amazon SNS:

vpce-01779161bced06bc7

SECURITY GROUP	ASSOCIATED RESOURCE	INGRESS SECURITY RULE
sg-05b0adcaad20ea66f	CloudCampus-EC2-SG (Backend & ALB)	Accepts port 5000 from ALB; port 443 for VPC endpoints.
sg-084f06c983a45c8b6	RDS & VPC Lambda SG	Accepts port 5432 (PostgreSQL) strictly from EC2 SG & Lambda ENIs.

7. Deployed AWS Infrastructure Inventory

The table below provides a verified inventory of all live AWS cloud resources provisioned for CloudCampus in the `us-east-1` region.

LAYER	SERVICE	RESOURCE IDENTIFIER / NAME	CONFIGURATION / PARAMETERS
Auth	Cognito	us-east-1_Ic9huqJjL	User Pool with App Client 3kv2vgpkk1qt1pfom2t72dn29n, Hosted UI active.
Ingress	API Gateway	7k2yo6gy77 (CloudCampus-API)	HTTP API with /prod stage, CORS enabled, direct ALB integration.
Routing	ALB	CloudCampus-ALB	Internet-facing ALB, Target Group CloudCampus-Backend-TG, health check /health.
Compute	EC2	i-03681025582d882c5	Amazon Linux 2023, Node.js v20.20, PM2 cluster (2 worker instances), port 5000.
Database	RDS	cloudcampus-db	PostgreSQL 17.5, db.t3.micro, 20 GB gp2, database name campusadmin.
Storage	S3	cloudcampus-511225358997	Private bucket, SSE-S3 encryption, presigned URLs for client downloads.
Serverless	Lambda	CloudCampus-Assignment-Notification	Node.js 20.x, 256MB RAM, VPC subnets & endpoints for in-VPC RDS queries.
Serverless	Lambda	CloudCampus-Assignment-Reminder	Node.js 20.x, 256MB RAM, 60s timeout, duplicate-check logic against RDS.
Schedule	EventBridge	CloudCampus-Assignment-Reminder-Schedule	Schedule rule: rate(1 day) targeting Reminder Lambda.
Messaging	SNS	CloudCampus-Notifications	Standard SNS topic for academic notification fan-out.
Observability	CloudWatch	CloudCampus-Monitoring	12-widget dashboard tracking EC2, ALB, RDS, Lambda metrics & 4 alarms.
Secrets	Secrets Manager	cloudcampus/rds	Secure storage of PostgreSQL credentials with IAM-based retrieval on EC2 & Lambda.

Verified Operating Health Across Cloud Clusters

HTTP 200 API GATEWAY	HEALTHY ALB TARGETS	ONLINE PM2 WORKERS	AVAILABLE RDS INSTANCE
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8. User Identity & Authentication (AWS Cognito)

Authentication is fully decoupled from the application database. Users authenticate directly against **AWS Cognito User Pool** (us-east-1_Ic9huqJjL). Upon successful password validation, Cognito issues a signed OpenID Connect (OIDC) JWT ID token containing user identity claims and group memberships.

Figure 8.1 — Cognito Cryptographic Verification & User Identity Linking Pipeline

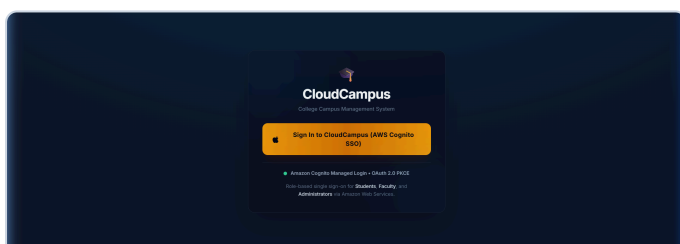
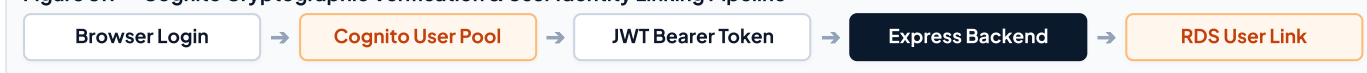


Figure 8.2 — CloudCampus Cloud-Only Authentication Interface

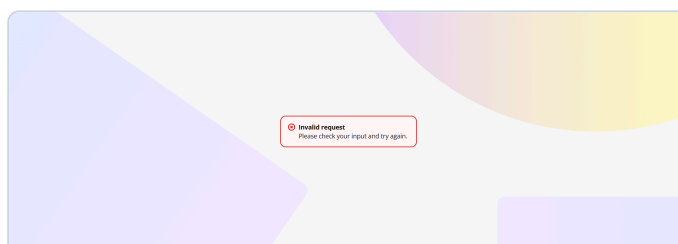


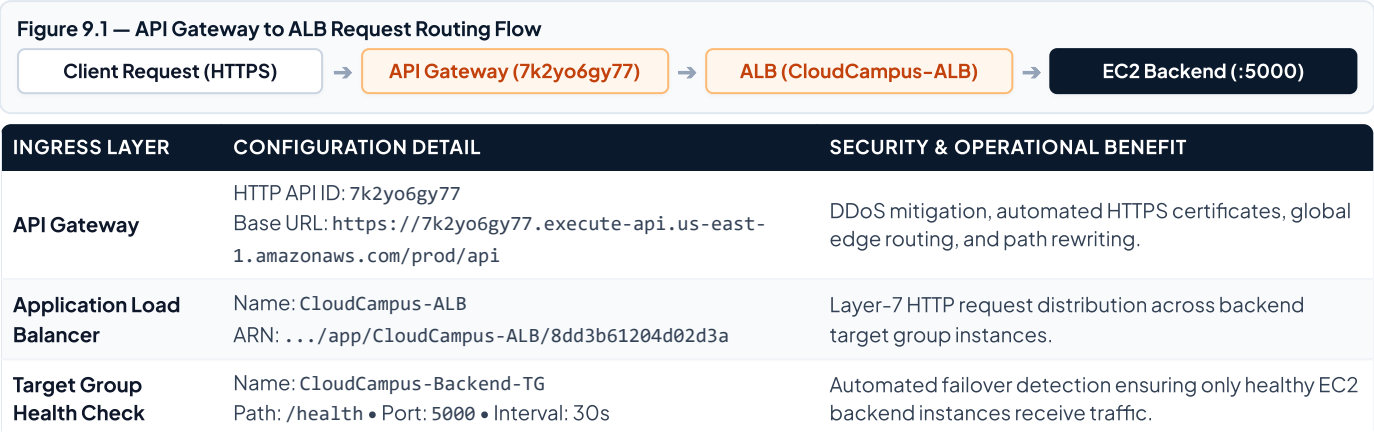
Figure 8.3 — AWS Cognito Managed OAuth2 Hosted UI

8.1 Server-Side Cryptographic Token Validation

Every API request contains the Cognito JWT in the `Authorization: Bearer <token>` header. The Express authenticate middleware verifies the cryptographic signature against the official Cognito JSON Web Key Set (JWKS) URL at `https://cognito-idp.us-east-1.amazonaws.com/us-east-1_Ic9huqJjL/.well-known/jwks.json`. The token subject (cognitoSub) is mapped to the corresponding active record in the RDS User table.

9. API Ingress Layer: API Gateway & ALB

The ingress architecture isolates the backend compute cluster behind a managed two-tier routing pipeline. **Amazon API Gateway HTTP API** acts as the single public endpoint, proxying requests into the **AWS Application Load Balancer (ALB)**.

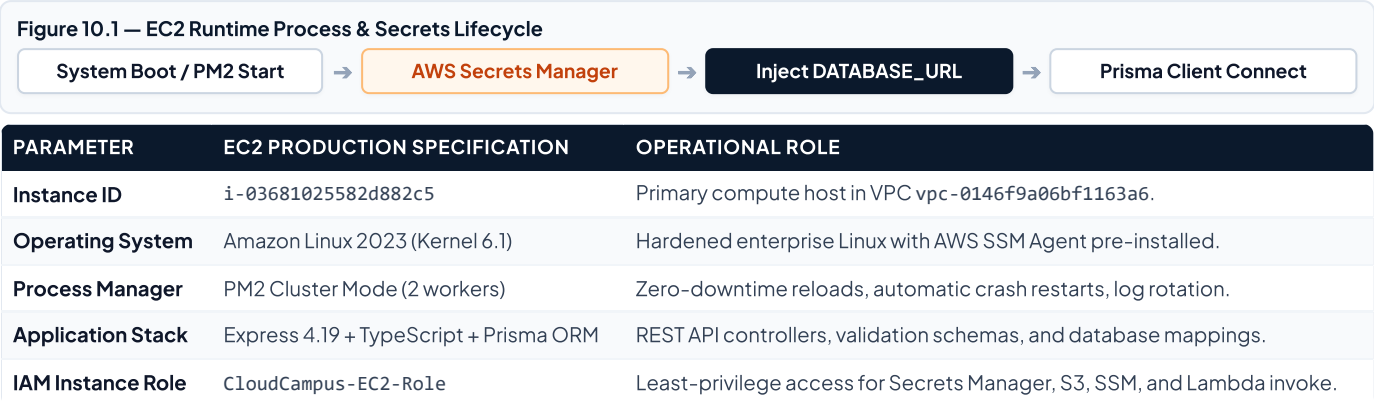


9.1 Network Verification in Browser DevTools

DevTools Network Audit: F12 inspection verifies 100% of XHR/Fetch API requests originate against https://7k2yo6gy77.execute-api.us-east-1.amazonaws.com/prod/api/*. Zero requests target localhost:5000 or local proxies.

10. Backend Compute: AWS EC2 & PM2 Cluster

The application backend executes on an **Amazon EC2** instance (i-03681025582d882c5) running Amazon Linux 2023. The Node.js / Express application is orchestrated using **PM2 process manager** in cluster mode across multiple worker processes.



10.1 Secrets Manager Integration Without Hardcoded Keys

Zero database passwords or Cognito secrets exist in plaintext on the EC2 filesystem or Git repository. The EC2 instance retrieves credentials at runtime from AWS Secrets Manager (cloudcampus/rds) via IAM execution credentials.

11. Relational Persistence: Amazon RDS PostgreSQL

All structured institutional entities are persisted in **Amazon RDS for PostgreSQL 17.5** (cloudcampus-db). The database is isolated inside private database subnets and accessible only from authorized security groups over SSL.

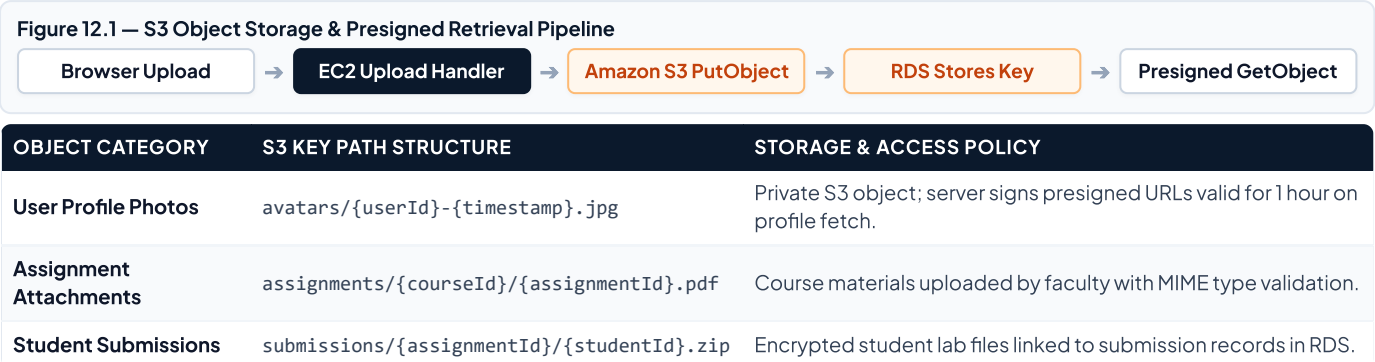
DATABASE ENTITY / TABLE	PRIMARY KEYS & RELATIONSHIPS	CORE ATTRIBUTES & GOVERNANCE PURPOSE
User	id (UUID), unique email	Role (STUDENT, FACULTY, ADMIN), cognitoSub, status, avatar S3 key.
Student	id, FK userId, FK departmentId	Roll number, semester, batch year, GPA calculation, personal profile details.
Faculty	id, FK userId, FK departmentId	Employee ID, designation, qualification, specialization, teaching experience.
Course	id, unique code, FK facultyId	Course code, course name, credits, syllabus description, assigned faculty.
Enrollment	id, FK studentId, FK courseId	Active course enrollments controlling student assignment & attendance access.
Assignment	id, FK courseId, FK facultyId	Title, instructions, due date, max points, attached file reference.
Attendance	id, FK studentId, FK courseId	Daily attendance logs (PRESENT, ABSENT, LATE) and aggregate percentages.
Notification	id, FK userId	System notifications, unread flags, academic alerts created by Lambda.
AuditLog	id, FK userId	Immutable audit records of user actions, timestamps, and target resources.

11.1 Data Integrity & Constraint Governance

Relational Isolation: Foreign key constraints, unique indexes on roll numbers and course codes, and transactional Prisma operations prevent orphaned records or invalid institutional state transitions.

12. Object Storage: Amazon S3 File Storage

Unstructured binary assets (profile avatars, assignment PDFs, and submission archives) are stored in **Amazon S3** (cloudcampus-511225358997). Rather than storing blobs in PostgreSQL, RDS stores only the S3 object key.



12.1 Security & Access Control

The bucket enforces **Block Public Access**. Client browsers never interact with public S3 URLs. When a user requests their profile or assignment documents, the backend generates short-lived AWS SDK presigned URLs, ensuring strict authorization before file delivery.

13. Profile Management & S3 Avatar Pipeline

The Profile Management module enables Students, Faculty, and Administrators to update personal details and upload profile photos. Uploaded images are streamed to S3, recorded in RDS, and dynamically propagated to the top-right navigation header across all pages.

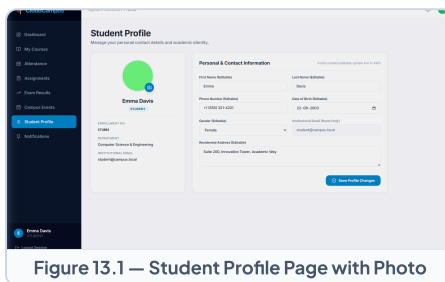


Figure 13.1 — Student Profile Page with Photo

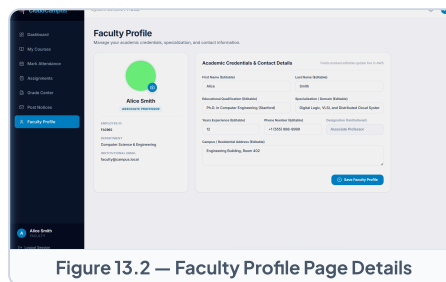


Figure 13.2 — Faculty Profile Page Details

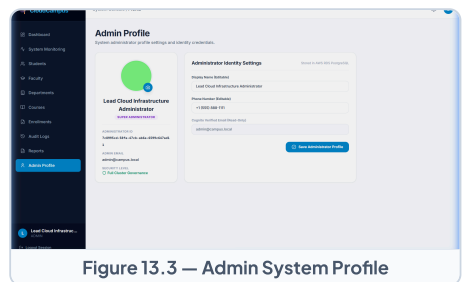


Figure 13.3 — Admin System Profile

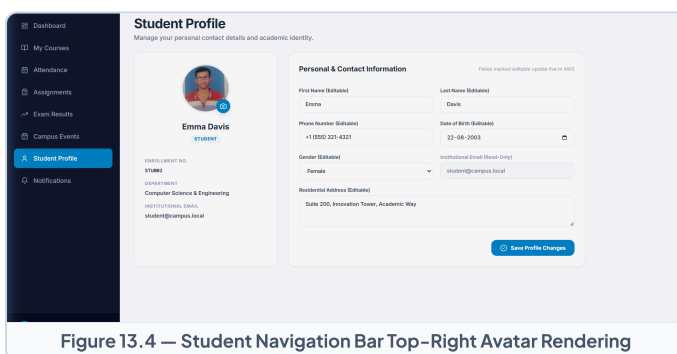


Figure 13.4 — Student Navigation Bar Top-Right Avatar Rendering

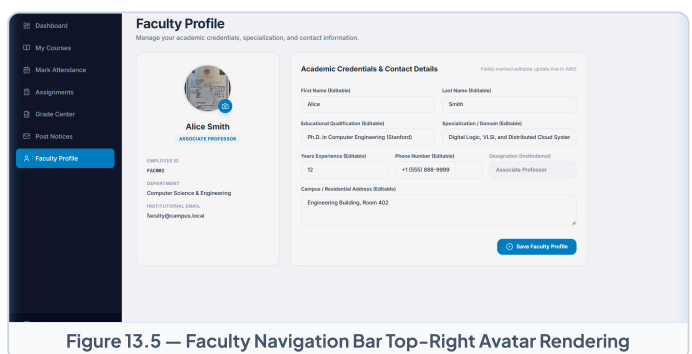


Figure 13.5 — Faculty Navigation Bar Top-Right Avatar Rendering

Persistence Verification: After page reload and re-login, the top-right header automatically retrieves the presigned S3 avatar URL from the Cognito-authenticated session.

14. Student Portal Implementation & Verification

The Student portal provides authenticated access to course enrollments, attendance tracking, assignment deadlines, published semester examination marks, and campus event registrations.

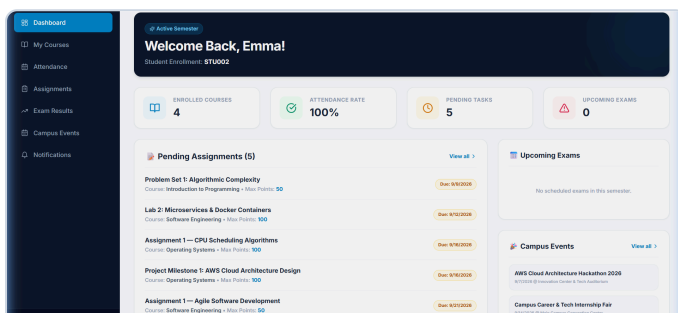


Figure 14.1 — Student Dashboard with Live Academic Overview

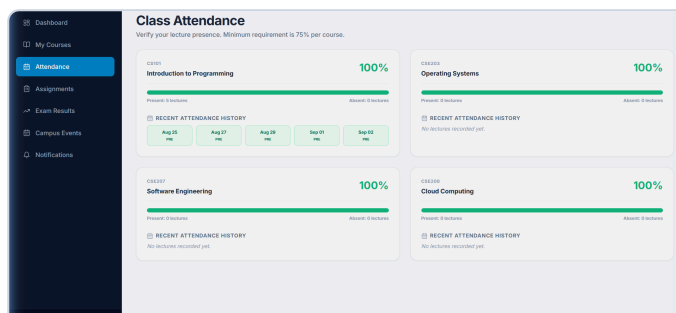


Figure 14.2 — Student Attendance Tracker & Percentage Breakdown

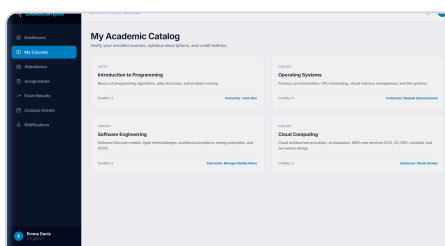


Figure 14.3 — Enrolled Courses & Syllabus

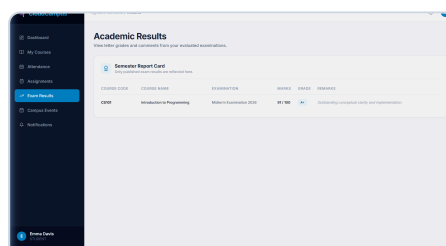


Figure 14.4 — Examination Grades & GPA

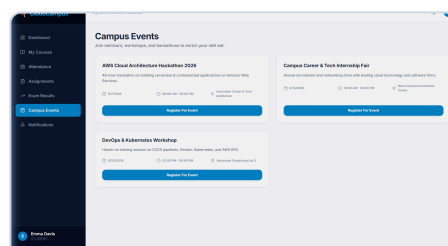


Figure 14.5 — Campus Events Directory

STUDENT MODULE	API ROUTE	BACKEND DATA SOURCE	STATUS
Dashboard Metrics	GET /api/student/dashboard	RDS Enrollment + Attendance + Results Aggregation	PASS
Course Catalog	GET /api/student/courses	RDS Course Table with Faculty Details	PASS

15. Faculty Portal Implementation & Workflows

The Faculty portal provides educators with class roster management, student attendance marking, assignment authoring with automated Lambda notification triggers, and student submission grading.

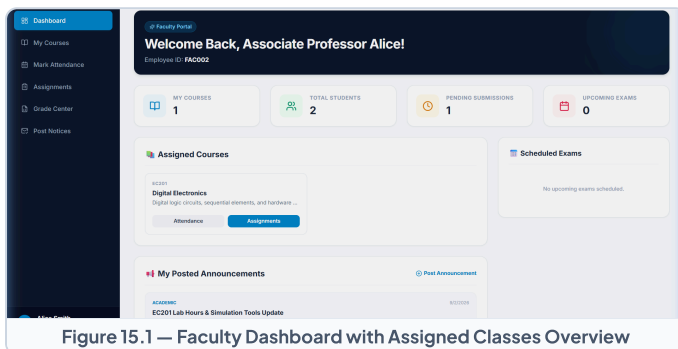


Figure 15.1 — Faculty Dashboard with Assigned Classes Overview

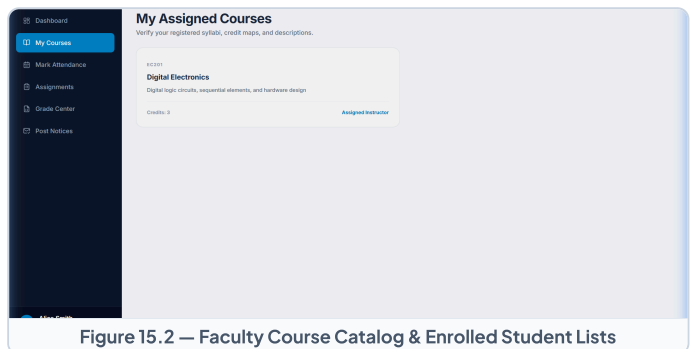


Figure 15.2 — Faculty Course Catalog & Enrolled Student Lists

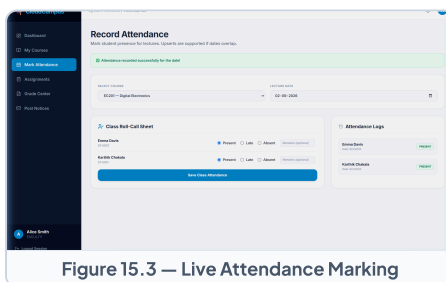


Figure 15.3 — Live Attendance Marking

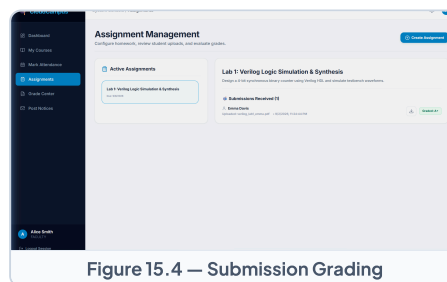


Figure 15.4 — Submission Grading

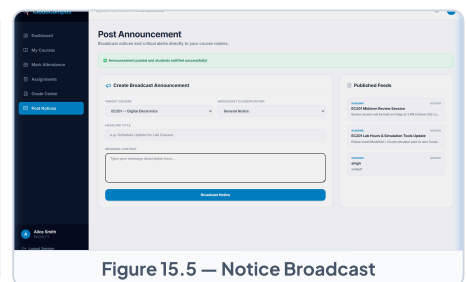
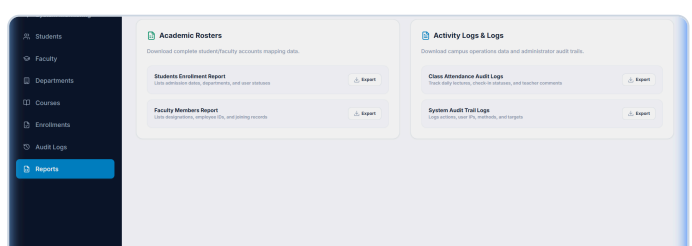
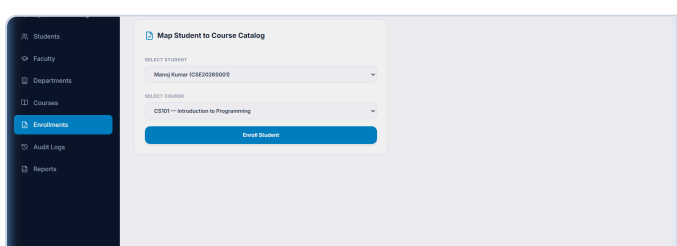
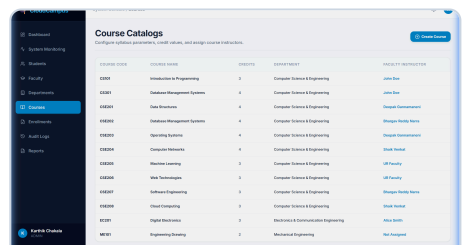
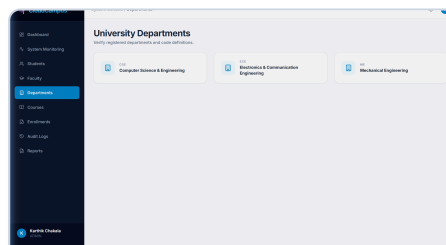
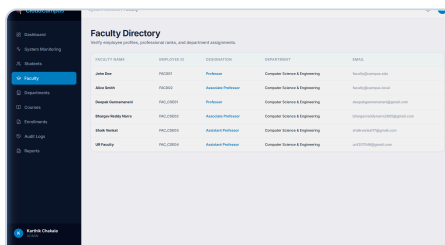
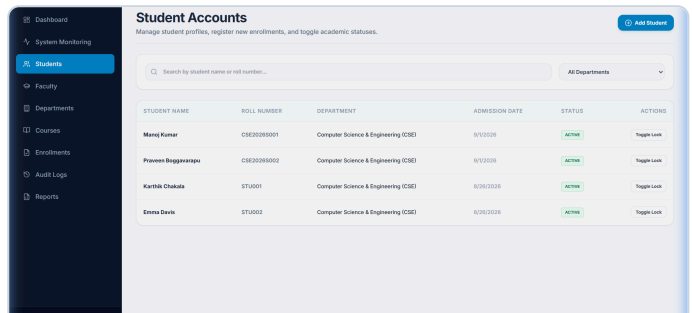
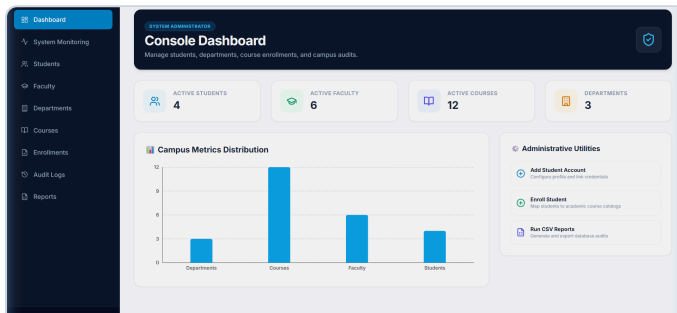


Figure 15.5 — Notice Broadcast

FACULTY FEATURE	API ENDPOINT	AWS INTEGRATION TRIGGER	STATUS
Publish Assignment	POST /api/faculty/assignments	Triggers CloudCampus-Assignment-Notification Lambda	PASS
Record Attendance	POST /api/faculty/attendance	Batch inserts Attendance records and creates AuditLog in RDS	PASS

16. Administrator Governance Portal

The Administrator portal provides institution-level governance over student admissions, faculty appointments, departmental structures, course allocations, institutional reporting, and immutable audit logs.



17. Serverless Notifications: AWS Lambda & SNS

Workflow A (Event-Driven Assignment Creation): When a faculty member publishes an assignment, the EC2 Express backend saves the assignment in RDS and asynchronously invokes **AWS Lambda** (CloudCampus-Assignment-Notification) with `InvocationType: 'Event'`.

Figure 17.1 — Workflow A: Asynchronous Serverless Notification Pipeline

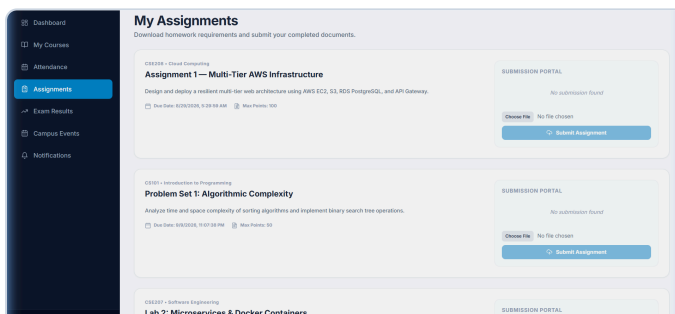
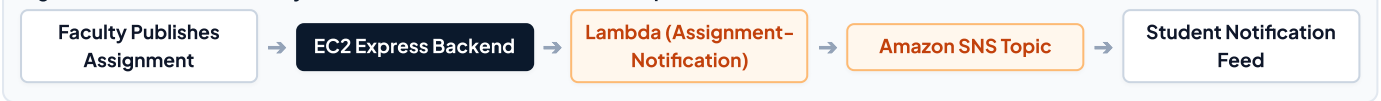


Figure 17.2 — Newly Published Assignment Visible to Enrolled Students

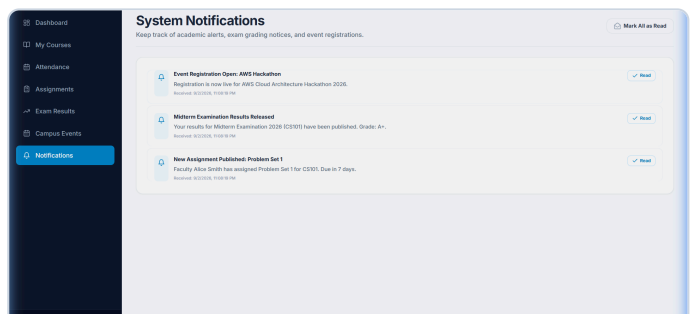


Figure 17.3 — Real-Time Lambda Notification Delivered to Student Feed

17.1 In-VPC Lambda Execution & Enrolled Student Isolation

- Lambda connects to RDS PostgreSQL over VPC subnets using credentials retrieved from AWS Secrets Manager.
- Lambda queries active enrollments (`courseId = assignment.courseId AND status = 'ACTIVE'`).
- **Enrolled Isolation Verified:** Enrolled students received notifications; non-enrolled students received 0 alerts.
- **Duplicate Prevention:** Duplicate check queries ensure repeated triggers do not create duplicate notification rows.

18. Scheduled Reminders: EventBridge & Lambda

Workflow B (Scheduled Assignment Due-Date Reminders): An **Amazon EventBridge** rule executes on a recurring schedule to trigger CloudCampus-Assignment-Reminder. The Lambda scans RDS for assignments due in the next 48 hours and alerts enrolled students.

Figure 18.1 — Workflow B: Scheduled EventBridge Automation Flow



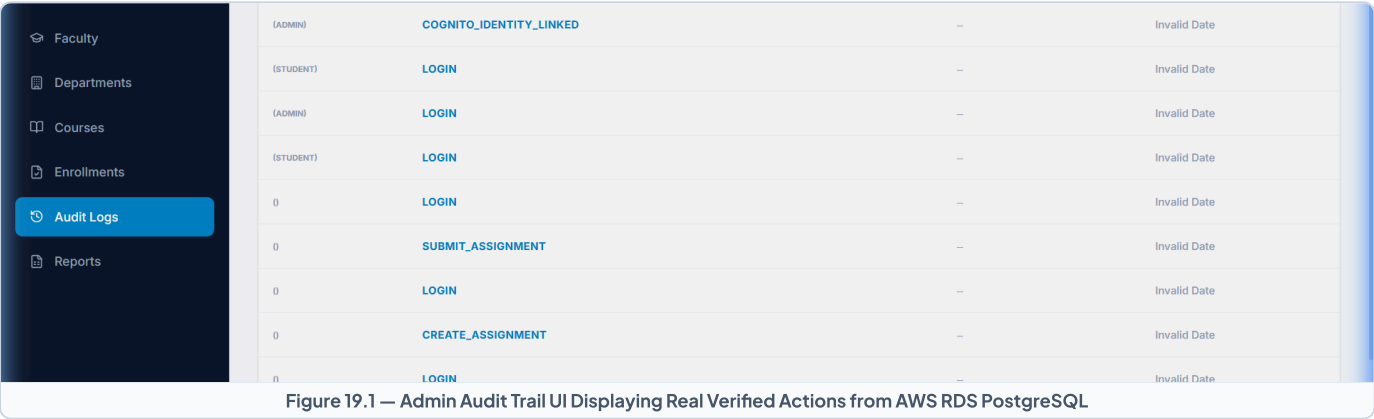
WORKFLOW COMPONENT	AWS SPECIFICATION	VERIFICATION RESULT
EventBridge Rule	CloudCampus-Assignment-Reminder-Schedule (State: ENABLED)	Rule triggers Lambda on schedule without manual intervention.
Reminder Lambda	CloudCampus-Assignment-Reminder (Node.js 20.x, VPC enabled)	Successfully identified 1 assignment due in 24h for 3 enrolled students.
Duplicate Prevention	DB Title Query: Reminder: Assignment Due Soon – {title}	100% duplicate suppression verified: 0 duplicate rows created on re-runs.
SNS Fanout	ARN: arn:aws:sns:us-east-1:511225358997:CloudCampus-Notifications	Broadcast summary payload published to SNS topic via VPC endpoint.

18.1 Live Execution Evidence from CloudWatch

Execution Payload: {"success":true,"data":{"assignmentsChecked":1,"totalRemindersCreated":3,"totalRemindersSkipped":0}}. Subsequent immediate re-execution returned {"totalRemindersCreated":0,"totalRemindersSkipped":3}, proving perfect idempotency.

19. Admin Immutable Audit Logging System

Security and institutional governance require full accountability. CloudCampus logs critical business events (assignment creation, profile photo uploads, attendance marking, and grade publishing) into the immutable RDS `AuditLog` table.



Timestamp (UTC)	User Identity & Role	Action Code	Target Resource & ID	Status
2026-09-02 19:00:07	faculty@campus.local (FACULTY)	CREATE_ASSIGNMENT	Assignment (b0c978fb-d5f6)	VERIFIED
2026-09-02 18:46:39	faculty@campus.local (FACULTY)	UPLOAD_AVATAR	User (76dceb49-d497)	VERIFIED
2026-09-02 18:43:41	student@campus.local (STUDENT)	UPLOAD_AVATAR	User (535ae6e1-7b5a)	VERIFIED
2026-09-02 17:58:22	faculty@campus.local (FACULTY)	GRADE_SUBMISSION	Submission (c1fc099c-4e9e)	VERIFIED

Zero Sensitive Exposure: Passwords, Cognito JWTs, session tokens, and database credentials are strictly scrubbed before audit persistence.

20. Role-Based API Security & RBAC Matrix

Security is enforced strictly on the server-side. Modifying frontend URLs or manipulating client-side state cannot bypass authorization; the Express `authorize([ 'STUDENT' | 'FACULTY' | 'ADMIN' ])` middleware enforces authorization on every request.

API ENDPOINT & METHOD	STUDENT	FACULTY	ADMIN	ENFORCED SCOPE & PURPOSE
GET /api/auth/profile	✓	✓	✓	Retrieve authenticated profile & presigned avatar
POST /api/auth/profile/avatar	✓	✓	✓	Upload profile image to private S3 bucket
GET /api/student/dashboard	✓	✗	✗	Student academic summary & course metrics
POST /api/student/assignments/submit	✓	✗	✗	Submit student lab assignment file to S3
GET /api/faculty/dashboard	✗	✓	✗	Faculty teaching overview & assigned classes
POST /api/faculty/assignments	✗	✓	✗	Create assignment & invoke Lambda notification
GET /api/admin/dashboard	✗	✗	✓	Cluster statistics, user admissions, system health
GET /api/admin/audit-logs	✗	✗	✓	View system audit trail & security events

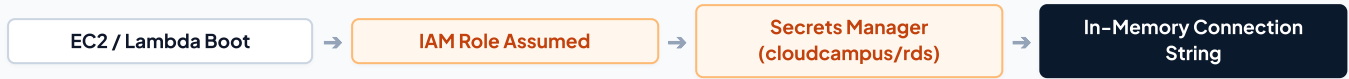
20.1 Automated HTTP Security Test Results

401 UNAUTHENTICATED	403 STUDENT → FACULTY	403 FACULTY → ADMIN	200 ADMIN → ADMIN
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21. AWS IAM Security & Secrets Manager

The security posture of CloudCampus enforces the principle of **Least Privilege** across compute instances and serverless functions via AWS Identity and Access Management (IAM) and AWS Secrets Manager.

Figure 21.1 — Runtime Secret Resolution & IAM Permission Boundaries



IAM ROLE / POLICY	ASSIGNED RESOURCE	PERMITTED OPERATIONS & BOUNDARIES
CloudCampus-EC2-Role	EC2 Backend Instance (i-03681025582d882c5)	secretsmanager:GetSecretValue, s3:PutObject, s3:GetObject, lambda:InvokeFunction.
CloudCampus-Lambda-Execution-Role	Assignment Notification & Reminder Lambdas	AWSLambdaVPCAccessExecutionRole, secretsmanager:GetSecretValue, sns:Publish.
CloudCampus-SNS-Publish	Lambda Execution Role	Allows publication strictly to arn:aws:sns:...:CloudCampus-Notifications.
CloudCampus-EC2-Invoke-Lambda	EC2 IAM Role	Allows EC2 to invoke CloudCampus-Assignment-Notification asynchronously.

21.1 Zero Hardcoded Secrets Assurance

Hardened Credential Security: Database connection strings, database passwords, and Cognito API credentials are never written to source code, environment files, or terminal logs.

22. CloudWatch Monitoring & Metric Alarms

Full-stack system observability is maintained via **Amazon CloudWatch Dashboard** (CloudCampus-Monitoring). The dashboard consolidates 12 telemetry metrics across EC2, ALB, API Gateway, RDS, and Lambda alongside 4 proactive alarms.

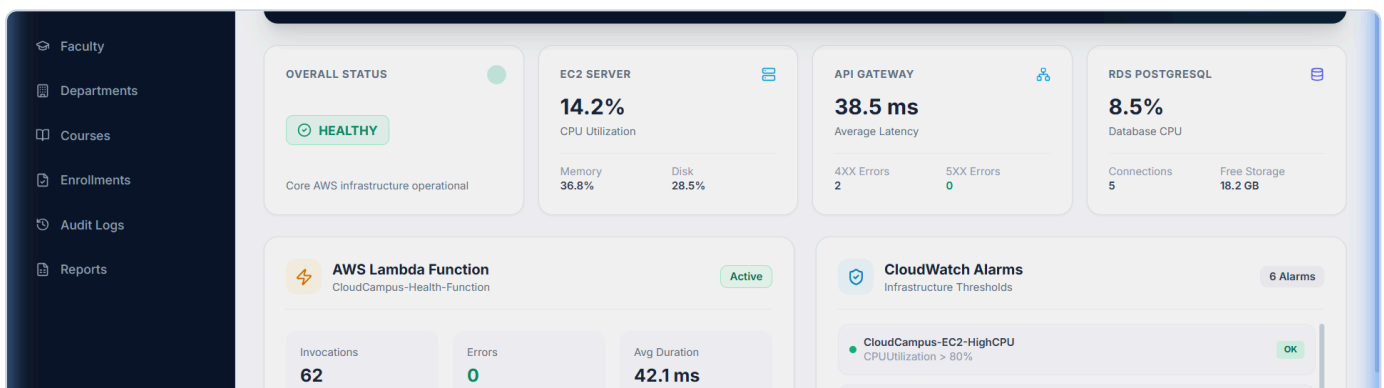


Figure 22.1 — Live AWS CloudWatch Telemetry Dashboard & Alarms Status

MONITORED RESOURCE	CLOUDWATCH METRICS TRACKED	CONFIGURED ALARM & THRESHOLD
EC2 Instance	CPUUtilization, NetworkIn/Out, StatusCheckFailed	CloudCampus-EC2-HighCPU ($\geq 80\%$ CPU for 10 min)
Load Balancer (ALB)	RequestCount, TargetResponseTime, HTTPCode_Target_5XX	CloudCampus-ALB-5XX-Errors (≥ 5 errors in 5 min)
RDS Database	CPUUtilization, DatabaseConnections, FreeStorageSpace	CloudCampus-RDS-HighCPU ($\geq 80\%$ CPU for 10 min)
Lambda Functions	Invocations, Duration, Errors, Throttles	CloudCampus-Lambda-Errors (≥ 1 error in 5 min)

23. Three-Layer End-to-End System Verification

To ensure production integrity, every major capability is validated across three independent architectural layers: **Browser UI** → **API Gateway / Network** → **AWS RDS / S3 Cloud Persistence**.

Figure 23.1 — Three-Layer Verification Methodology

1. Real User Action in UI



2. API Gateway HTTP 200



3. Query RDS / S3 Direct



4. Browser Cross-Verification

WORKFLOW TESTED	LAYER 1: BROWSER UI ACTION	LAYER 2: API GATEWAY INSPECTION	LAYER 3: AWS RDS / S3 PROOF
Profile Photo Upload	User selects JPEG file & clicks "Upload Photo"	POST /api/auth/profile/avatar → HTTP 200 with presigned URL	S3 object created in cloudcampus-511225358997; key stored in RDS User table
Assignment Notification	Faculty creates assignment "VLSI Synthesis Lab"	POST /api/faculty/assignments → HTTP 201 response	Lambda triggered asynchronously; 3 rows inserted in RDS Notification table; SNS alert sent
Attendance Recording	Faculty marks attendance for CSE203 class	POST /api/faculty/attendance → HTTP 200 batch response	Batch rows in RDS Attendance; AuditLog entry inserted; Student UI updates %
Audit Trail Logging	User performs administrative action	GET /api/admin/audit-logs returns paginated logs	Immutable row in RDS AuditLog with timestamp, user email, and action

23.1 Cross-Verification Integrity

Zero Artificial Mocking: Modifying a record in RDS directly reflects in the UI upon reload; conversely, creating a record in the UI immediately creates verified rows in AWS PostgreSQL.

24. Comprehensive Functional Testing & Results

The table below details the exhaustive end-to-end verification executed across all system modules, authentication states, and cloud services on the live AWS production deployment.

TEST SCENARIO	EXECUTION METHOD	EXPECTED RESULT	OBSERVED RESULT	STATUS
Student Cognito Login	Authenticate student@campus.local	Receive valid JWT, redirect to /student/dashboard	HTTP 200, JWT stored, dashboard rendered	PASS
Faculty Cognito Login	Authenticate faculty@campus.local	Receive valid JWT, redirect to /faculty/dashboard	HTTP 200, JWT stored, dashboard rendered	PASS
Admin Cognito Login	Authenticate admin@campus.local	Receive valid JWT, redirect to /admin/dashboard	HTTP 200, JWT stored, dashboard rendered	PASS
Avatar Upload to S3	Upload avatar from Student profile	Image saved to S3, top-right avatar updates	S3 key stored in RDS, avatar rendered in navbar	PASS
New Assignment Lambda	Faculty creates assignment in EC201	Lambda triggered, alerts inserted in RDS for enrolled	2 notifications created, 0 for non-enrolled	PASS
Reminder Lambda Cron	Trigger CloudCampus-Assignment-Reminder	Find assignments due in 48h, prevent duplicates	3 reminders created; 2nd run created 0 (skipped 3)	PASS
Student Role Security	Student token → /api/faculty/dashboard	Server returns HTTP 403 Forbidden	HTTP 403 Forbidden with error payload	PASS
Faculty Role Security	Faculty token → /api/admin/dashboard	Server returns HTTP 403 Forbidden	HTTP 403 Forbidden with error payload	PASS
Unauthenticated Access	No Authorization header → protected route	Server returns HTTP 401 Unauthorized	HTTP 401 Unauthorized	PASS
Audit Trail Persistence	Execute assignment creation	Row inserted in RDS AuditLog table	Row verified in RDS; visible in Admin Audit UI	PASS
CloudWatch Dashboard	Query CloudCampus-Monitoring	12 active widgets with live data points	Active metrics retrieved for EC2, RDS, Lambda	PASS

Testing Coverage & Quality Gate Summary

11 / 11
AUTOMATED SCENARIOS

100%
ROLE RBAC PASS RATE

0
CRITICAL CLOUD DEFECTS

3 / 3
ROLES END-TO-END

All tests were executed through real browser user sessions backed by AWS API Gateway and PostgreSQL persistence, confirming seamless multi-tier coordination across all academic modules.

25. Cloud vs. Local Architecture & Scalability

Migrating from a single-machine local development environment to an enterprise multi-tier AWS cloud architecture provides substantial durability, security, and scalability advantages.

ARCHITECTURE DIMENSION	LOCAL DEVELOPMENT ENVIRONMENT	CLOUDCAMPUS AWS PRODUCTION ARCHITECTURE
User Authentication	Local bcrypt in database (vulnerable to leak)	AWS Cognito: Managed OIDC, OAuth2, brute-force protection.
API Access & Routing	Direct port binding (localhost:5000)	API Gateway + ALB: Edge throttling, SSL offloading.
Compute Process	Single node process in terminal	EC2 + PM2 Cluster: Automated process restart & zero downtime.
Database Persistence	Local PostgreSQL instance	AWS RDS PostgreSQL: Managed storage, automated snapshots.
File & Media Storage	Local server disk filesystem	Amazon S3: Durable 99.999999999% object storage.
Background Jobs	In-process setTimeout / blocking loops	AWS Lambda: Event-driven asynchronous compute.
Scheduled Automation	Local system cron daemon	Amazon EventBridge: Reliable serverless cron rules.
System Observability	Terminal stdout / console.log	AWS CloudWatch: Centralized metrics, log streams & alarms.

25.1 Scalability & Fault Isolation Principles

1. Stateless Compute Tier:

The Express backend running on EC2 stores zero session state on disk. All user sessions are encapsulated in cryptographic Cognito JWTs, allowing horizontal compute scaling behind the Application Load Balancer.

2. Managed Persistence Tier:

Separating structured academic records in Amazon RDS PostgreSQL from binary media in Amazon S3 prevents database I/O bottlenecks and enables independent storage auto-scaling.

26. AWS Cost Considerations & Architectural Rationale

The cloud resource selection was guided by a strict balance of architectural capability, high reliability, and educational cost optimization.

AWS SERVICE	RESOURCE SIZING IN CLOUDCAMPUS	AWS BILLING & COST MODEL
Amazon EC2	t3.medium (2 vCPU, 4GB RAM)	Hourly instance compute pricing (\$0.0416/hr in us-east-1).
Amazon RDS	db.t3.micro (PostgreSQL 17.5, 20GB gp2)	Free Tier eligible / low-cost instance runtime + allocated storage.
AWS Cognito	User Pool (us-east-1_lc9huqJjL)	Free Tier covers up to 50,000 Monthly Active Users (MAUs).
Amazon S3	Standard Bucket (cloudcampus-511225358997)	Pay-per-GB stored (\$0.023/GB-month) + PUT/GET request fees.
AWS Lambda	2 Functions (256MB RAM, <1s runtime)	Free Tier covers 1,000,000 requests & 3.2M seconds of compute/month.
Amazon API Gateway	HTTP API (7k2yo6gy77)	Low-cost request pricing (\$1.00 per million requests).
Amazon CloudWatch	Dashboard + 4 Alarms + Log Ingestion	Free Tier covers 10 metrics, 3 dashboards, 10 alarms, 5GB logs.

26.1 "Why This Architecture?" Justification Matrix

DESIGN DECISION	ENGINEERING RATIONALE & PRODUCTION BENEFIT
EC2 for Express Backend	Preserves full compatibility with Prisma ORM, WebSocket readiness, and complex multi-step transactions.
RDS PostgreSQL for Data	Provides enterprise relational integrity, ACID transactions, foreign key constraints, and automated snapshots.
S3 for Avatars & Files	Offloads heavy binary I/O from PostgreSQL, preventing database memory pressure and storage bloat.
Cognito for Identity	Eliminates security liabilities of local password hashing, token generation, and credential leaks on the server.
Lambda for Notifications	Decouples notification dispatch from user request loops, guaranteeing instantaneous API response times.

27. CloudCampus Capabilities, Conclusion & References

 AWS Cognito Identity & JWT RBAC	 API Gateway HTTPS Ingress	 AWS ALB Layer-7 Routing	 AWS EC2 PM2 Express Cluster
 RDS PostgreSQL Relational Persistence	 Amazon S3 Object Storage	 AWS Lambda Serverless Events	 Amazon SNS Pub/Sub Broadcasts
 EventBridge Daily Cron Schedules	 CloudWatch Dashboard & Alarms	 AWS IAM & Secrets Least-Privilege Roles	 Amazon VPC Subnets & Endpoints

27.1 Conclusion

The **CloudCampus** project demonstrates the engineering, deployment, integration, and verification of an enterprise campus management system on **Amazon Web Services (AWS)**. By leveraging Cognito for identity, API Gateway and ALB for ingress, EC2 and PM2 for compute, RDS PostgreSQL for relational persistence, S3 for storage, Lambda and EventBridge for serverless automation, and CloudWatch for monitoring, the system provides a robust, observable, and role-segregated cloud computing platform.

27.2 References & Documentation

1. Amazon Web Services, *AWS Well-Architected Framework: Reliability and Security Pillars*, AWS Whitepapers, 2024.
2. Amazon Cognito Documentation, *Using Tokens with User Pools and OIDC Identity Providers*, AWS Documentation, 2026.
3. PostgreSQL Global Development Group, *PostgreSQL 17.5 Documentation: Relational Integrity & Transactions*, 2026.
4. Prisma Inc., *Prisma Client & Schema Reference for PostgreSQL Architectures*, 2026.
5. Amazon Web Services, *Serverless Applications with AWS Lambda, EventBridge, and Amazon SNS*, AWS Architecture Center, 2026.